

AI HOME DESIGNER
IMPLEMENTATION BLUEPRINT
PART 1

STATUS:
PRODUCTION IMPLEMENTATION

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1. REPOSITORY STRUCTURE

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ai-home-designer/

apps/
├── web/
├── api/
└── admin/

packages/
├── ui/
├── types/
├── sdk/
├── config/
└── utils/

services/
├── ai/
├── exporter/
└── worker/

infrastructure/
├── docker/
├── kubernetes/
└── terraform/

docs/

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2. DEVELOPMENT ORDER

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STEP 1

Infrastructure

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STEP 2

Database

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STEP 3

Authentication

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STEP 4

REST API

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STEP 5

Frontend

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STEP 6

2D Editor

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STEP 7

AI Engine

↓

STEP 8

3D Engine

↓

STEP 9

Testing

↓

STEP 10

Production

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3. CODING STANDARDS

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Language

TypeScript

Indentation

2 Spaces

Strict Mode

Enabled

ESLint

Required

Prettier

Required

100% Typed APIs

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4. BRANCH STRATEGY

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main

develop

feature/*

bugfix/*

release/*

hotfix/*

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5. RELEASE STRATEGY

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Semantic Versioning

v1.0.0

v1.1.0

v2.0.0

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END OF IMPLEMENTATION BLUEPRINT PART 1

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AI HOME DESIGNER

IMPLEMENTATION BLUEPRINT

PART 2

STATUS:

PRODUCTION IMPLEMENTATION STANDARDS

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6. BACKEND STRUCTURE

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apps/api/

src/

modules/

- |— auth/
- |— users/
- |— organizations/
- |— projects/
- |— rooms/

- walls/
- doors/
- windows/
- roofs/
- stairs/
- materials/
- furniture/
- ai/
- export/
- import/
- notifications/
- admin/

common/

config/

middleware/

guards/

interceptors/

filters/

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7. FRONTEND STRUCTURE

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apps/web/

app/

components/

features/

hooks/

stores/

services/

pages/

styles/

assets/

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8. DATABASE MIGRATIONS

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Each database change requires

Migration File

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Rollback File

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Schema Validation

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Automatic Tests

Migration naming

001_initial_schema

002_users

003_projects

004_rooms

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9. CODE REVIEW RULES

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Every Pull Request requires

✓ Successful Build

- ✓ All Tests Passed
- ✓ Lint Passed
- ✓ Type Check Passed
- ✓ Security Scan Passed
- ✓ One Reviewer Approval

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10. DEFINITION OF DONE

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A feature is complete only if

- ✓ Code implemented
- ✓ Tests written
- ✓ Documentation updated
- ✓ API documented
- ✓ Performance verified
- ✓ Security reviewed
- ✓ No critical bugs

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11. RELEASE CHECKLIST

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Before each release

- ✓ Database backup
- ✓ Migration test
- ✓ Load test
- ✓ Security scan

- ✓ API validation
- ✓ Frontend build
- ✓ AI regression test
- ✓ Monitoring active

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12. LONG-TERM MAINTENANCE

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- Monthly
- Dependency updates
 - Security patches
 - Performance review
 - Database optimization

- Quarterly
- Architecture review
 - AI model evaluation
 - Infrastructure audit

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13. IMPLEMENTATION STATUS

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- Architecture Complete
- Specifications Complete
- Implementation Blueprint .. Complete

NEXT STEP

Actual source code development.

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END OF IMPLEMENTATION BLUEPRINT
IMPLEMENTATION PLANNING COMPLETE
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```

```
# AI HOME DESIGNER
# DEVELOPER HANDBOOK
# PART 1
```

STATUS:
DEVELOPMENT EXECUTION GUIDE

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1. DEVELOPMENT PRINCIPLES
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Every implementation must follow:

- Single Responsibility Principle (SRP)
- Open/Closed Principle (OCP)
- Liskov Substitution Principle (LSP)
- Interface Segregation Principle (ISP)
- Dependency Inversion Principle (DIP)

Architecture Style

Modular Monolith (MVP)

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Microservices (Enterprise)

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2. GENERAL RULES

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Every function

- Typed
 - Documented
 - Tested
 - Reusable
 - Side-effect free whenever possible
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No duplicated business logic.

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3. API RULES

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Every endpoint must have

- ✓ Request DTO
 - ✓ Response DTO
 - ✓ Validation
 - ✓ Authorization
 - ✓ Logging
 - ✓ Error handling
 - ✓ OpenAPI documentation
- =====
- =====

4. DATABASE RULES

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Never

- Delete production data permanently
- Break foreign keys
- Bypass migrations

Always

- Use transactions
- Use indexes where appropriate
- Validate constraints

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5. AI RULES

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The AI must never:

- Return invalid geometry
- Ignore hard constraints
- Produce non-schema-compliant JSON
- Modify user-locked elements
- Hide validation errors

Every AI result must pass validation before it is presented to the user.

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6. TESTING POLICY

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Minimum requirements

Unit Tests >90% coverage

Integration Tests Required

End-to-End Tests Required

Regression Tests Required

Performance Tests Required

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7. DOCUMENTATION POLICY

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Every public module must include

- Purpose
- Inputs
- Outputs
- Examples
- Error cases
- Performance notes

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8. FINAL EXECUTION ORDER

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1. Infrastructure

2. Database

3. Authentication

4. Backend API

5. Frontend

6. 2D Editor

7. AI Engine

8. 3D Engine

9. Testing

10. Production Deployment

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END OF DEVELOPER HANDBOOK PART 1

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AI HOME DESIGNER
MASTER IMPLEMENTATION CHECKLIST

STATUS:
FINAL EXECUTION CHECKLIST

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1. PROJECT INITIALIZATION

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- ☐ Git Repository
- ☐ Monorepo Structure
- ☐ Branch Protection
- ☐ CI/CD
- ☐ Docker
- ☐ Environment Variables
- ☐ Secrets Management
- ☐ Documentation

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2. DATABASE

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- PostgreSQL
- Extensions
- Initial Schema
- Migrations
- Seed Data
- Indexes
- Constraints
- Backup
- Monitoring

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3. BACKEND

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- Authentication
- Authorization
- User Module
- Organization Module
- Project Module
- Room Module
- Wall Module
- Door Module
- Window Module
- Roof Module
- Material Module
- Furniture Module

- AI Module
- Export Module
- Import Module
- Notifications
- Admin

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4. FRONTEND

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- Authentication UI
- Dashboard
- Project List
- Project Editor
- Canvas
- Toolbar
- Property Panel
- Material Library
- AI Assistant
- Export UI
- Settings
- Profile

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5. AI

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- Planner

- ☐ Geometry
- ☐ Validator
- ☐ Optimizer
- ☐ Cost
- ☐ Energy
- ☐ Interior
- ☐ Explanation
- ☐ Prompt Library
- ☐ AI Memory

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6. 3D

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- ☐ Viewer
- ☐ Rendering
- ☐ Materials
- ☐ Lighting
- ☐ Camera
- ☐ Walk Mode
- ☐ Section Mode
- ☐ Export

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7. IMPORT / EXPORT

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- ☐ PDF

☐ PNG

☐ SVG

☐ DXF

☐ IFC

☐ JSON

☐ GLTF

☐ OBJ

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8. SECURITY

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☐ JWT

☐ RBAC

☐ Encryption

☐ HTTPS

☐ Rate Limits

☐ Audit Logs

☐ Monitoring

☐ Security Scan

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9. TESTING

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☐ Unit Tests

☐ Integration Tests

☐ E2E Tests

- ☐ Load Tests
- ☐ Stress Tests
- ☐ Security Tests
- ☐ Regression Tests

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10. DEPLOYMENT

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- ☐ Docker
- ☐ Kubernetes
- ☐ Monitoring
- ☐ Logging
- ☐ Alerts
- ☐ CDN
- ☐ Auto Scaling
- ☐ Backups

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11. COMMERCIAL

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- ☐ Landing Page
- ☐ Pricing
- ☐ Billing
- ☐ Subscription
- ☐ Documentation
- ☐ Analytics

☐ Customer Support

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12. VERSION 1.0 RELEASE

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Release Criteria

- ✓ Stable
- ✓ Secure
- ✓ Tested
- ✓ Documented
- ✓ Scalable
- ✓ Production Ready

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PROJECT STATUS

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- Architecture ✓
- Specifications ✓
- Implementation Plan ✓
- Execution Checklist ✓
- Ready for Development ✓

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END OF MASTER IMPLEMENTATION CHECKLIST

END OF COMPLETE PROJECT DOCUMENTATION

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